

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

## PROFESSIONAL EXPERIENCE

### Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and captured schematics for **optical control circuits**, electromechanical systems, and analogue signal chain processing, integrating TIAs and DACs for enhanced performance.
- Performed **PCB layout** for complex electronic designs, including exposure to high-speed RF layout principles and impedance matching for critical signal paths.
- Supported component selection for **space-grade hardware**, referencing AS/EN9100 and JEDEC standards for aerospace deployment requirements.
- Developed prototype **subsystems** from initial concept through to aerospace-qualified product stages, working within strict mechanical and thermal constraints.
- Created detailed **power budgets** for subsystems, ensuring efficient energy management and adherence to stringent aerospace power specifications.
- Simulated **PCBs** for next-generation Free Space Optics (FSO) transceivers using SPICE, developing noise models and link budgets to validate performance.

## SKILLS

**Electronics Design** – Analogue Circuit Design, Digital Circuit Design, Schematic Capture, PCB Layout, Signal Integrity, Impedance Matching, SPICE Simulation, Noise Modelling, Link Budgets

**Aerospace Standards** – AS/EN9100, JEDEC Standards, Space-Grade Hardware, Aerospace Deployment

**Communications Protocols** – CAN, SPI, UART, I2C

**Optical Systems** – Free Space Optics (FSO), Optical Control Circuits, Optical Readout, Signal Chain Processing, TIAs, DACs

**Software & Tools** – Altium, EAGLE, KiCAD, PADs, Linux

**Laboratory Equipment** – Oscilloscopes, Multimeters, Spectrum Analyzers

## PROJECTS

### Space-Grade FSO Transceiver Simulation & Design (HCL Technologies)

- Modeled analogue readout and signal processing circuits, including transimpedance amplifiers (TIAs) and digital-to-analogue converters (DACs), for an FSO transceiver system.
- Executed SPICE simulations to verify circuit performance and developed noise models, contributing to the overall link budget calculations for the optical communication system.
- Documented design specifications and simulation results, providing clear traceability for the development of aerospace-qualified electronic components.

### Aerospace Control System Prototyping (HCL Technologies)

- Designed and captured schematics for electromechanical and optical control circuits, adhering to AS/EN9100 principles for high-reliability applications.
- Managed component selection process, prioritizing parts suitable for harsh aerospace environments and long-term operational stability.
- Collaborated with mechanical engineers to ensure designs met stringent thermal and physical constraints for integration into a constrained space platform.

## EDUCATION

### M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

### B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

## CERTIFICATIONS

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- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)